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Oefeningenreeks 6E nr. 8.1

$\sin x = T_n(\sin x, 0)(x) + R_n(x)$

$R_n(x) = \dfrac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(c)$

Stel $-|x| < c < |x|$:

$R_{2n+1}(x) = \dfrac{x^{2n+2} \sin^{(2n+2)} c}{(2n+2)!}$

$\overset{x=-0.7}{\Rightarrow} |R_{2n+1}(-0.7)| = \dfrac{0.7^{2n+2} \sin^{(2n+2)} c}{(2n+2)!}$

$\overset{|\sin^{(k)} c| \leq 1}{\Rightarrow} |R_{2n+1}(-0.7)| \leq \dfrac{0.7^{2n+2}}{(2n+2)!}$

$\overset{n=4}{\Rightarrow} |R_9(-0.7)| \leq \dfrac{0.7^{10}}{10!}$

$\dfrac{0.7^{10}}{10!} \approx 7.78 \cdot 10^{-9} < 5 \cdot 10^{-8}$

We nemen dus de Taylorpolynoom tot en met graad 9:

$T_n(\sin x, 0)x = \sum_{k=0}^{n} \dfrac{(\sin 0)^{(k)}}{k!} x^k$

$= \dfrac{\sin 0}{1} x^0 + \dfrac{\cos 0}{1} x + \dfrac{-\sin 0}{2} x^2 + \dfrac{-\cos 0}{6} x^3 + \dfrac{\sin 0}{24} x^4 + \dfrac{\cos 0}{120} x^5 + \dfrac{-\sin 0}{720} x^6 + \dfrac{-\cos 0}{5040} x^7 + \dfrac{\sin 0}{40320} x^8 + \dfrac{\cos 0}{362880} x^9 ...$

$= x + -\dfrac{x^3}{6} + \dfrac{x^5}{120} - \dfrac{x^7}{5040} + \dfrac{x^9}{362880} ...$

$\sin(-0.7) \approx -0.7 + -\dfrac{(-0.7)^3}{6} + \dfrac{(-0.7)^5}{120} - \dfrac{(-0.7)^7}{5040} + \dfrac{(-0.7)^9}{362880}$

$\Rightarrow \sin(-0.7) \approx -0.6442177$

References